

Simon Lejoly

PhD student in Machine Learning · University of Namur

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ABOUT

I'm a PhD student working on **probabilistic machine learning**. I build models that know what they don't know, write the software that makes them *fast*, and create *fancy animations* to make them easy to understand. Currently focused on probabilistic models for *planetary science* data.

EDUCATION

2023 – now **PhD in Machine Learning** · *HuMaLearn Lab — UNamur*
Explored the modeling capabilities of Gaussian processes on Venusian atmospheric data

2021 – 2023 **Data Science Master** · *University of Namur*
Presented my master thesis on "Personalised Speech Compression using Deep Learning"

2018 – 2021 **Computer Science Bachelor** · *University of Namur*
Finished my general informatics cursus with *great distinction*

2000 – 2018 **Homeschooling** · *Well... at home!*
Spent my childhood being curious and learning by myself (and with my parents)

PUBLICATIONS

2026 **Probabilistic detection of outliers in Venusian atmospheric profiles from VEx/SOIR**
S. Lejoly, A. Piccialli, A. Mahieux, A. C. Vandaele, B. Frénay — *EuroPlanet Science Congress 2026*

2026 **Towards Probabilistic Modelling of Venus Express/SOIR's Acousto-Optic Tunable Filter**
A. Gunessee, A. Mahieux, S. Robert, A. Piccialli, I. Thomas, **S. Lejoly**, V. Delchevalerie, A. C. Vandaele, B. Frénay — *Copernicus GmbH*

2024 **MAGMA Gaussian Processes to complete Venusian atmospheric profiles from SOIR**
S. Lejoly, A. Piccialli, A. Mahieux, A. C. Vandaele, B. Frénay — *EuroPlanet Science Congress 2024*

2024 **Le Projet VAMOS**
A. Piccialli, J. Sermeus, **S. Lejoly**, S. Claes, J. Stepanovic, A. Mahieux, K.Lefever, S. Fratta et A. C. Vandaele — *Science Connection n°71*

2024 **Extrapolating Venusian Atmospheric Profiles using MAGMA Gaussian Processes**
S. Lejoly, A. Piccialli, A. Mahieux, A. C. Vandaele, B. Frénay — *ESANN 2024 proceesdings*

TALKS & POSTERS

Oct 2026 **Examples of use of MIMOSA, a probabilistic framework for functional data analysis** — *BeNeLearn 2026*· Poster

Talk **Bayesian Machine Learning at scale for functional data analysis** — *Europlanet Machine Learning Summer School*

Talk **MAGMA Gaussian Processes to complete Venusian atmospheric profiles from SOIR** — *Europlanet Science Congress*

Talk **Probabilistic detection of outliers in Venusian atmospheric profiles from VEx/SOIR** — *Europlanet Science Congress*

Poster **Extrapolating Venusian atmospheric profiles using MAGMA Gaussian Processes** — *ESANN*

Talk **Multi-Task Gaussian Processes Applied to Spatial Aeronomy Data** — *University of Antwerp*

Talk **Les modèles probabilistes pour l'analyse, la prédiction et le clustering de données atmosphériques** — *UNamur — Advanced Deep Learning course*

SOFTWARE

mimosa-ml ★ 19
MIMOSA: a fully-featured multi-task Gaussian process framework for analysing functional data.

kernax-ml ★ 9
Kernax: a Python package providing efficient mathematical kernel implementations for probabilistic machine learning models, built with JAX.

manim-presentations
A Python framework for creating composable manim-slides presentations.

eqxbatch
Batched evaluation of Equinox modules: a vmap that lives in the PyTree.

KiloGuide ★ 2
A minimal, get-started guide for kilobots, with various tutorials.

MISC

🎤 **MT180** — National finalist of "My Thesis in 180 Seconds".

⛺ **BACV trainer** — I train future holiday camp leaders.

🥁 **Drums** — I slam sticks on stuff.

🎮 **E-sport** — I organised e-sport tournaments.

♞ **Half decent at Chess** — 1350 Elo in blitz.

☕ **Coffee** — Certified coffee enthusiast.

Animated version: simlej18.github.io/resume